# Jay Collins

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## Education

**University of Colorado Boulder**, Boulder, CO
*B.S. Electrical Engineering* Expected May 2027
GPA 3.84, Dean's List

## Technical Courses

- Circuits I & II (A), Linear Systems (A), Electronics Design Lab (A–), Embedded Software Engineering (A)
- Data Structures (A), Programmable Digital Systems (A), Microelectronics (A), Fields and Waves (A–)
- Applications of Embedded Systems (A), Embedded AI (A), Digital Logic (A), Probability and Statistics (A)

## Skills

**Languages:** C, C++, Python, RISC-V Assembly, Verilog

**Protocols & Interfaces:** SPI, I2C, UART, I2S, PWM, GPIO Interrupts, Bluetooth, WiFi, REST APIs

**Software & Debugging:** FreeRTOS, STM32CubeIDE, GDB, UART/Serial Logging, Renode, PyTorch, ExecuTorch, GitHub

**Hardware & Testing:** PCB Design (Altium), SMD, THT, Soldering, Oscilloscopes, Function Generators, DMM, Power Supplies

**Core Engineering Skills:** Circuit Design, Embedded Systems, Digital Logic, State Machines, IoT

## Projects

**Wake-Word Detection on STM32 – Embedded ML**

- Built a real-time keyword spotter in C++ on an STM32F407. It runs a 4-class CNN through ExecuTorch on 16 kHz PDM microphone audio, using STFT features and a 1 MB flash budget.
- Trained with time-shift, gain, and noise augmentation and added a tuned voice-activity detector, then validated the system in Renode simulation and on hardware.

**World's Worst Alarm Clock – WiFi-Connected Embedded System**

- Built an ESP32 alarm clock in C++ with a state machine, 30 Hz SPI OLED, I2C RTC with NTP sync, and weather via REST API.
- Moved networking into a FreeRTOS task on the second core, using task notifications and a mutex so the alarm never stalls.
- Took the hardware from breadboard to soldered protoboard to a custom 3D-printed enclosure.

**ShapeShifter Hybrid Synthesizer – Senior Capstone (In Progress)** Sep 2026 – Apr 2027

- Project manager for a 6-person team building a synthesizer for an industry client on a Daisy Seed3 (STM32H750).
- Built the full breadboard prototype, wiring 32 panel controls into a C++ audio engine with oscillators and a granular sampler.
- As software manager, split the firmware into a real-time audio callback and a lower-priority control loop that reads four 12-bit MCP3208 SPI ADCs and drives preset-recall LEDs through MCP23017 I2C expanders.

## Experience

**Engineering Learning Assistant – CU Boulder** Jan 2026 – Present

- Learning Assistant for Linear Systems and Exploring ECE, supporting 150+ students in weekly office hours.
- Explained complex concepts like convolution and Fourier transforms in clear, accessible language.

**Deckhand – Hawai'i Nautical** May 2024 – Present

- Crew member aboard a 54 ft sailing and snorkeling catamaran, handling maintenance and passenger safety during tours.
- Assisted with maintenance and monitoring of onboard electrical systems, applying hands-on problem-solving to identify and address issues before they impacted operations. Lifeguard and CPR certified.